



VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2008

H2 CHEMISTRY

9746/1

PAPER 1 Multiple Choice

Thursday
18/09/2008

1 hour
40 marks

Additional materials:

Data Booklet
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your *name*, *CT group* and *NRIC / FIN* on the answer sheet in the spaces provided.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

A Data Booklet is provided.

Any rough working should be done in this booklet.

This question paper consists of **18** printed pages.

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Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

- 1** To determine the percentage of nitrogen present in a snack, 1.0 g of the snack was boiled with concentrated sulphuric acid to convert all the nitrogen into ammonium sulphate. The ammonium salt obtained was then boiled with excess aqueous sodium hydroxide to liberate the ammonia which was passed into 25.0 cm³ of 0.20 mol dm⁻³ HCl. The unreacted HCl required 20.0 cm³ of 0.10 mol dm⁻³ NaOH for complete neutralisation.

What is the percentage by mass of nitrogen in the snack?

- A** 2.8%
B 4.2%
C 7.0%
D 8.4%
- 2** The hydroxylammonium ion, NH₃OH⁺, reduces Fe³⁺ to Fe²⁺ in acidic medium. If 2 mol of Fe³⁺ are reduced by 1 mol of NH₃OH⁺, which of the following nitrogen-containing species is formed in the reaction?

- A** NH₄⁺
B N₂O
C NO₂
D NO₃⁻

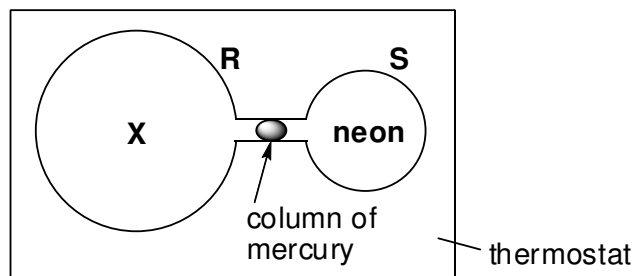
- 3** Two elements **X** and **Y** have the following properties.

- **X** and **Y** form ionic compounds Na₂**X** and Na₂**Y** respectively.
- Element **Y** forms **YF₆** molecule whereas **X** is not able to do so.

Which pair of electronic configurations of **X** and **Y** is correct?

	X	Y
A	[He]2s ² 2p ²	[Ne]3s ² 2p ⁴
B	[He]2s ² 2p ²	[Ne]3s ² 3p ⁴
C	[He]2s ² 2p ⁴	[Ne]3s ² 3p ²
D	[He]2s ² 2p ⁴	[Ne]3s ² 3p ⁴

- 4 The volume of bulb **R** is twice the volume of bulb **S**. **R** contains a gas **X** at the same pressure as the neon in **S**.



When the thermostat temperature is increased, which of the following gases in bulb **R** would cause the mercury column to move closer to bulb **S**?

- A** $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g}) \quad \Delta H > 0$
- B** $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g}) \quad \Delta H > 0$
- C** $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g}) \quad \Delta H < 0$
- D** $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) \quad \Delta H < 0$
- 5 Which of the following species is **not** planar?
- A** BF_3
- B** NO_3^-
- C** Al_2Cl_6
- D** COCl_2
- 6 The standard enthalpy change of formation of but-1-ene, $\text{C}_4\text{H}_8(\text{g})$, and ethene, $\text{C}_2\text{H}_4(\text{g})$, are x and $y \text{ kJ mol}^{-1}$ respectively. Calculate the standard enthalpy change of the reaction $2\text{C}_2\text{H}_4(\text{g}) \rightarrow \text{C}_4\text{H}_8(\text{g})$.

- A** $(x - 2y)$
- B** $(2y - x)$
- C** $\frac{1}{2}(x + y)$
- D** $\frac{1}{2}(y - x)$

- 7 The enthalpy change for the process represented by the equation



is equal to

- A** the second ionisation energy of magnesium.
B the enthalpy change of sublimation of magnesium.
C the sum of the first and second ionisation energies of magnesium.
D the sum of the first and second ionisation energies and the enthalpy change of atomisation of magnesium.
- 8 Some experimental data obtained from a reaction between **X**, **Y** and **Z** at 500K are shown below.

Experiment	Initial [X] / mol dm ⁻³	Initial [Y] / mol dm ⁻³	Initial [Z] / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	3.0	0.01	0.1	3 x 10 ⁻⁶
2	3.0	0.01	0.2	6 x 10 ⁻⁶
3	4.0	0.01	0.2	8 x 10 ⁻⁶
4	4.0	0.02	0.1	4 x 10 ⁻⁶

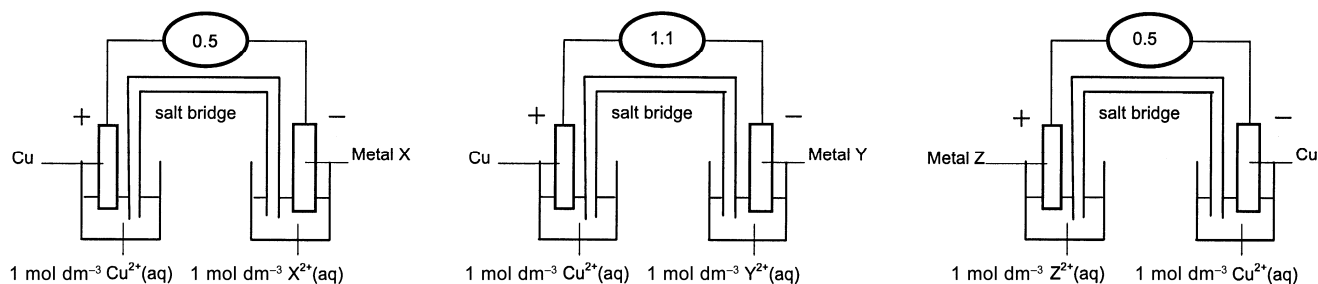
The rate equation for this reaction is

- A** rate = $k[\text{X}][\text{Y}]$
B rate = $k[\text{X}][\text{Z}]$
C rate = $k[\text{X}][\text{Y}][\text{Z}]$
D rate = $k[\text{X}][\text{Y}][\text{Z}]^2$
- 9 Two electrolytic cells containing $\text{P}_2(\text{SO}_4)_m(\text{aq})$ and $\text{Q}_2(\text{SO}_4)_n(\text{aq})$ are connected in series, where **P** and **Q** are both metals. The relative atomic masses of **P** to **Q** is 1 : 3. When a current is passed through the set-up, the ratio of the masses of **P** : **Q** liberated is 1 : 2. What are the values of m and n ?

	m	n
A	1	6
B	6	1
C	2	3
D	3	2

- 10 Use of the Data Booklet is relevant to this question.

Three electrochemical cells are set up as shown below. The *e.m.f* in volts is shown on each voltmeter.



These *e.m.f* indicate the order of reactivity of the metals. The order of the weakest to the strongest reducing agent is

- A X, Cu, Y, Z
 B Cu, Z, X, Y
 C Z, Cu, X, Y
 D Y, X, Z, Cu
- 11 For the reaction below, the value of the equilibrium constant, K_p , is 1.28 at 1000K.



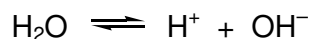
When equilibrium has been reached at 1000K, the partial pressure of **C** is found to be 0.15 atm. What is the initial pressure of **A**?

- A 0.43 atm
 B 0.32 atm
 C 0.26 atm
 D 0.13 atm

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- 12 Use of the Data Booklet is relevant to this question.

Water dissociates as shown.

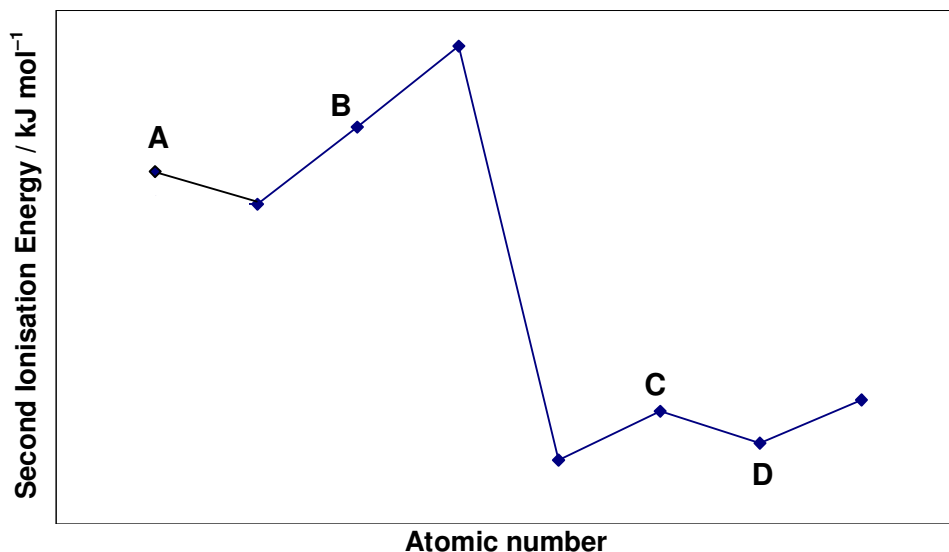


At 25°C, the equilibrium values of $[\text{H}^+] = 10^{-7} \text{ mol dm}^{-3}$ and $[\text{H}_2\text{O}] = \frac{1000}{18} \text{ mol dm}^{-3}$.

What is the order of increasing numerical value of pH, $\text{p}K_{\text{a}}$ and $\text{p}K_{\text{w}}$ for this equilibrium at this temperature? [$\text{p}K_{\text{w}} = -\log_{10} K_{\text{w}}$]

	smallest		largest
A	pH	$\text{p}K_{\text{w}}$	$\text{p}K_{\text{a}}$
B	pH	$\text{p}K_{\text{a}}$	$\text{p}K_{\text{w}}$
C	$\text{p}K_{\text{a}}$	$\text{p}K_{\text{w}}$	pH
D	$\text{p}K_{\text{w}}$	$\text{p}K_{\text{a}}$	pH

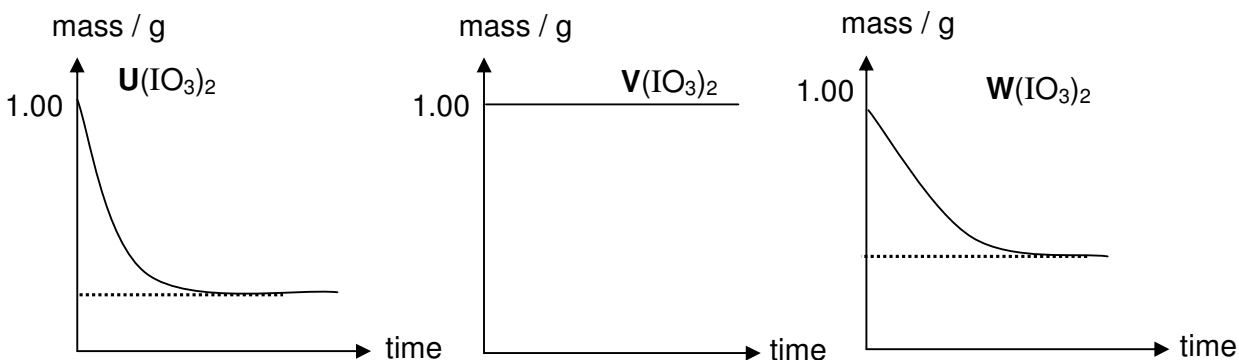
- 13 The following graph shows the second ionisation energy of eight consecutive elements.



Element **P** belongs to Period 3 of the Periodic Table. It has a high electrical conductivity and its chloride has a simple molecular structure.

With reference to the graph above, which of the options **A**, **B**, **C** or **D**, corresponds to the second ionisation energy of element **P**?

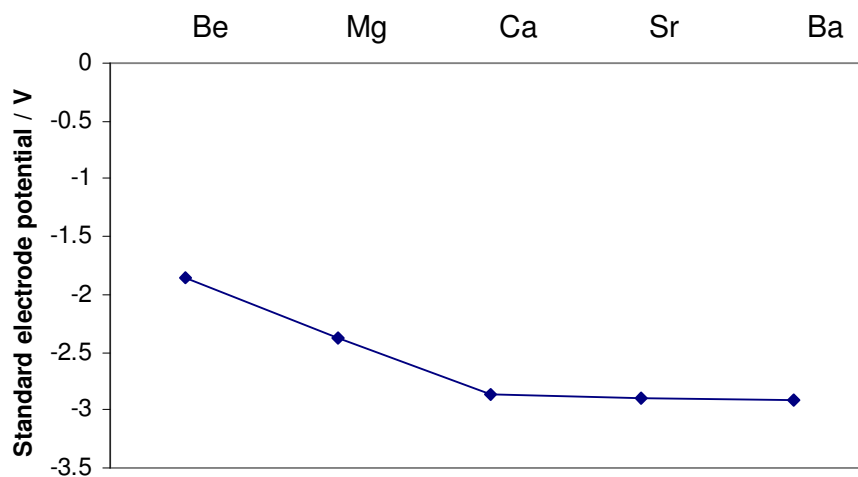
- 14 Which statement does **not** describe the chemical properties of Period 3 compounds?
- A SiO_2 is formed from the hydrolysis of SiCl_4 .
- B Al_2O_3 is soluble in both acid and base.
- C Na_2O and P_4O_{10} can be mixed in water to give an approximately neutral solution.
- D The acidity of Period 3 chlorides in water decreases from NaCl to PCl_5 .
- 15 Which statement is correct of the halogens and their compounds?
- A Silver iodide is soluble in concentrated ammonia.
- B When I_2 is shaken with aqueous NaBr , followed by the addition of hexane, a reddish-brown organic layer is obtained.
- C HCl decomposes to produce yellowish-green vapour on contact with a hot glass rod.
- D A violet vapour is produced when concentrated H_2SO_4 is added to solid sodium iodide.
- 16 **U**, **V** and **W** are Group II elements. The three graphs below show the change in mass when 1.00 g each of **U**(IO_3)₂, **V**(IO_3)₂ and **W**(IO_3)₂ were heated separately at a temperature of $T^\circ\text{C}$.



Which of the following shows the elements in order of increasing atomic number?

- A **U**, **W**, **V**
- B **V**, **W**, **U**
- C **V**, **U**, **W**
- D **W**, **U**, **V**

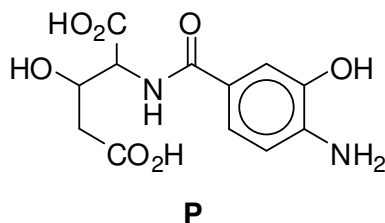
- 17 The following graph represents the trend in the standard electrode potential of Group II elements.



Which statement can be inferred from the above graph?

- A The elements form compounds with fixed oxidation state of +2.
 - B The reducing power of the elements increases down the group.
 - C Reactivity of the elements with water decreases down the group.
 - D The metals can be extracted from their ore by reduction with carbon.
- 18 Which solution will form a precipitate when aqueous silver nitrate is added?
- A $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$
 - B $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$
 - C $\text{Na}_3[\text{CrCl}_6]$
 - D $[\text{Cr}(\text{en})(\text{NH}_3)_2\text{Cl}_2]\text{NO}_3$

- 19 Compound **P** has the structure shown below.

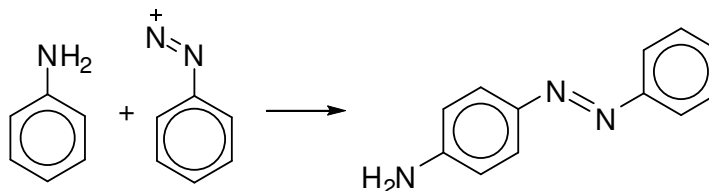


Which statement is **incorrect** about **P**?

- A** **P** exists as 4 stereoisomers.
- B** 1 mol of **P** reacts with 2 mol of dilute HCl on heating.
- C** 1 mol of **P** produces 4 mol of HCl on reaction with PCl_5 .
- D** **P** does not produce an alkaline gas when warmed with aqueous NaOH.
- 20 A typical protein forms several hundreds of hydrogen bonds and several thousands of van der Waals' forces in folding from primary to tertiary structures. Which thermodynamic state functions of the protein best represents the folding process?

	$\Delta H / \text{kJ mol}^{-1}$	$\Delta S / \text{J mol}^{-1} \text{K}^{-1}$
A	< 0	< 0
B	< 0	> 0
C	> 0	> 0
D	> 0	< 0

- 21 A bright yellow dye, aniline yellow can be synthesised from benzene in the following synthetic route.



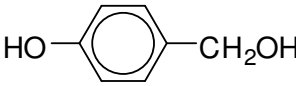
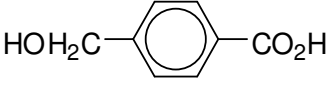
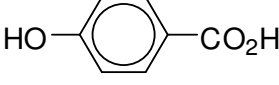
What is the type of reaction for the reaction shown above?

- A** electrophilic substitution
- B** electrophilic addition
- C** nucleophilic substitution
- D** nucleophilic addition

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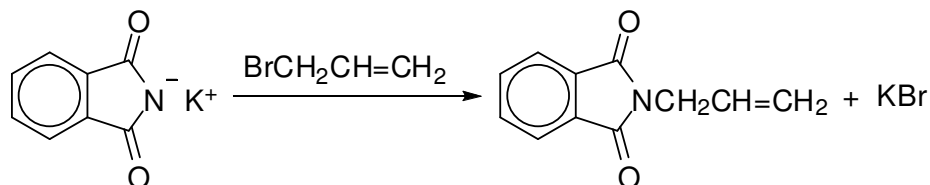
- 22 1 mol of an organic compound **X** reacts with 1 mol of CH_3COCl . 1 mol of **X** also reacts with PCl_5 to produce 2 mol of hydrogen chloride.

Which compound could be **X**?

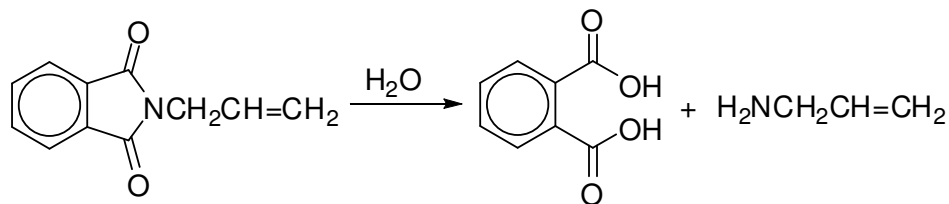
- A** 
- B** 
- C** 
- D** 

- 23 The Gabriel synthesis shown below gives a high yield of primary amines.

Step 1



Step 2



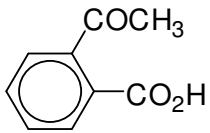
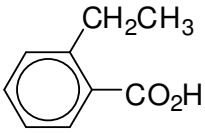
The type of reaction taking place in each of the steps is

- | | Step 1 | Step 2 |
|----------|----------------------------|------------|
| A | nucleophilic substitution | hydrolysis |
| B | electrophilic substitution | hydrolysis |
| C | nucleophilic substitution | oxidation |
| D | electrophilic substitution | oxidation |

24 Which compound **cannot** be synthesised from propan-1-ol within 2 steps or less?

- A propan-2-ol
- B ethanoic acid
- C propanoic acid
- D butanoic acid

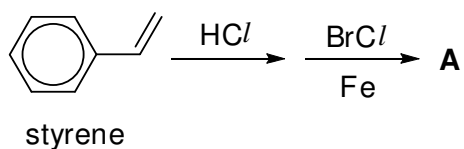
25 In which pair of molecules is the pK_a of molecule II smaller than that of molecule I?

- | | I | II |
|---|---|---|
| A |  |  |
| B | $C_6H_5NH_3^+$ | $C_6H_{13}NH_3^+$ |
| C | $CH_3CHClCH_2OH$ | $ClCH_2CH_2CH_2OH$ |
| D | $CH_3CONHCH_3$ | $H_3N^+CH(CH_3)CO_2H$ |

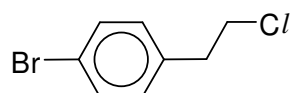
26 When aqueous bromine is added to phenol, the colour of bromine disappears and a white precipitate is formed. Which statement is correct?

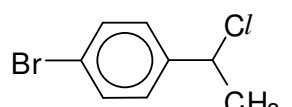
- A The solution becomes more acidic.
- B The mechanism is nucleophilic substitution.
- C The $-OH$ group of phenol is substituted by Br.
- D Bromine forms a colourless complex with phenol.

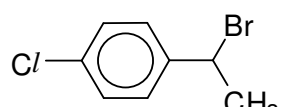
- 27 Styrene undergoes the following reactions to form compound **A**.

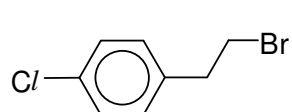


What is the structural formula of **A**?

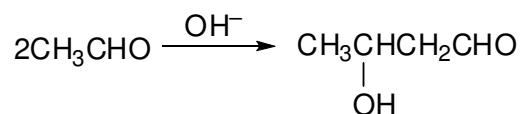
- A** 

B 

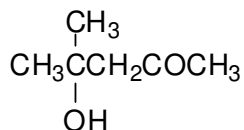
C 

D 

- 28 In the presence of dilute alkali, some aldehydes and ketones undergo the 'aldol' reaction' where they dimerise to form a hydroxycarbonyl compound (an aldol). For example, ethanal dimerises as shown to form 3-hydroxybutanal.



Which carbonyl compound will undergo the aldol reaction to produce the aldol shown below?



- A** CH_3COCH_3

B $\text{CH}_3\text{CH}_2\text{CHO}$

C $(\text{CH}_3)_2\text{CHCHO}$

D $\text{CH}_3\text{CH}_2\text{COCH}_3$

- 29 A halogenoalkane, **S**, gives an amine when heated in a sealed tube with excess ammonia. A nitrile, **T**, gives an amine by reacting it with LiAlH_4 in dry ether. Which pair of **S** and **T** will give the same amine after reaction?

	S	T
A	$\text{CH}_3\text{CH}_2\text{Cl}$	$\text{CH}_3\text{CH}_2\text{CN}$
B	$\text{CH}_3\text{CH}_2\text{CHClCH}_3$	$\text{CH}_3\text{CH}(\text{CN})\text{CH}_3$
C	$\text{CH}_3\text{CH}(\text{CH}_2\text{Cl})\text{CH}_3$	$\text{CH}_3\text{CH}(\text{CN})\text{CH}_3$
D	$\text{C}_6\text{H}_5\text{Cl}$	$\text{C}_6\text{H}_5\text{CN}$

- 30 Which reaction yields a carbon compound incorporating deuterium, D? [$\text{D} = {}^2_1\text{H}$]

- A** $\text{CH}_3\text{CH}_2\text{CN} \xrightarrow[\text{D}_2\text{O}]{\text{NaOD}}$
- B** $\text{CH}_3\text{CD}(\text{OD})\text{CO}_2\text{H} \xrightarrow[\text{heat}]{\text{acidified KMnO}_4}$
- C** $\text{CH}_3\text{COCl} \xrightarrow[\text{D}_2\text{O}]{\text{NaOD}}$
- D** $(\text{CH}_3)_3\text{COH} \xrightarrow{\text{conc D}_2\text{SO}_4}$

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which pairs of substances have similar types of bonding and structure?

- | | | |
|----------|---------------|-------------------------|
| 1 | BF_3 | AlF_3 |
| 2 | PH_3 | SiCl_3H |
| 3 | Diamond | SiC |

32 The value of the ionic product of water, K_w , varies with temperature.

temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
25	1.0×10^{-14}
62	1.0×10^{-13}

What can be deduced from this information?

- The ionic dissociation of water is an endothermic process.
- Water is a neutral liquid at 62°C.
- The association of water molecules by hydrogen bonding increases as temperature rises.

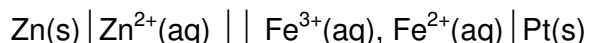
- 33 A given reaction has the overall equation and proceeds via the following mechanism shown below.

Overall equation: $2\text{A} + \text{B} \rightarrow \text{C} + \text{D}$

Mechanism: $\text{A} + \text{B} \rightarrow \text{C} + \text{E}$ (fast) $\Delta H = +45 \text{ kJ mol}^{-1}$
 $\text{A} + \text{E} \rightarrow \text{D}$ (slow) $\Delta H = -98 \text{ kJ mol}^{-1}$

Which statements are **incorrect**?

- 1 **E** is a catalyst in the reaction.
 - 2 The overall order of the reaction is 3.
 - 3 The enthalpy change for the overall reaction is -53 kJ mol^{-1} .
- 34 A cell with a standard potential, E°_{cell} , of $+1.53 \text{ V}$ is shown below using conventional notation.



Which of the following changes, when made separately to the set-up above, will increase the E°_{cell} ?

- 1 Adding aqueous iron(III) chloride to the $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell.
 - 2 Adding water to the Zn^{2+}/Zn half-cell.
 - 3 Adding water to the $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell.
- 35 An aqueous solution containing both potassium chloride and potassium iodide is treated with an excess of aqueous silver nitrate. The precipitate formed is filtered off and washed with distilled water. The precipitate is then shaken with aqueous ammonia and filtered off again.

Which ion is **not** present in the final filtrate?

- 1 chloride
- 2 iodide
- 3 potassium

The responses **A** to **D** should be selected on the basis of

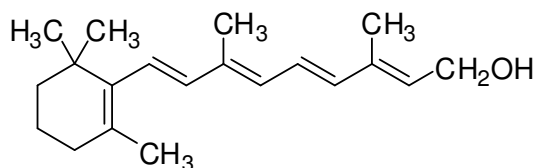
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

36 An element **G** has atomic number 22. Which compounds of **G** are possible?

- 1** **GS**
- 2** Na_3GF_6
- 3** GCl_4

37 Vitamin A is an important nutrient that can prevent night-blindness.

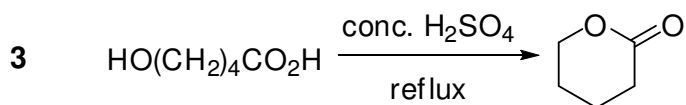
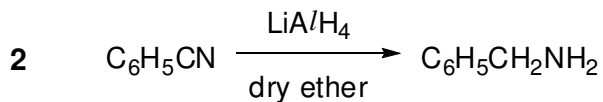
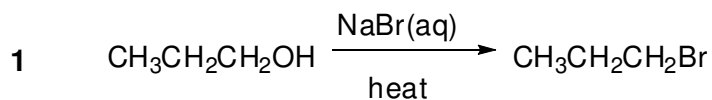


Vitamin A

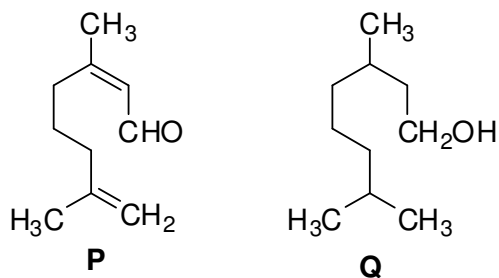
Which conclusions can be drawn from the structure of vitamin A?

- 1** It decolourises aqueous bromine.
- 2** It reacts with 6 mol of HCl under suitable conditions.
- 3** It has 32 stereoisomers.

38 Which synthesis would **not** yield the product shown?



39 The two compounds **P** and **Q** shown below are important flavours in citrus fruits. These compounds are commonly used in the food and perfume industries.



Which statements concerning **P** and **Q** are correct?

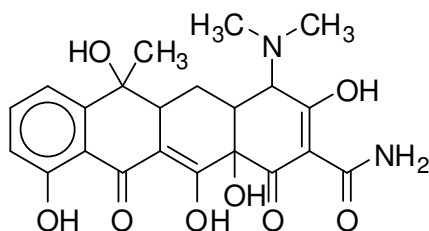
- 1 **Q** can be distinguished from **P** by using phosphorus pentachloride.
- 2 Reaction of **P** with hot acidified KMnO_4 gives two organic products.
- 3 **P** can be converted to **Q** using lithium aluminium hydride in dry ether.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

40 The anti-bacterial drug, tetracycline, has the structure shown below.



Tetracycline

Which reagents when added separately to tetracycline will liberate a gaseous product?

- 1** sodium
- 2** sodium hydroxide
- 3** sodium carbonate